CHEMICAL REACTION NETWORKS
5. Pathway analysis

Pathway Databases, Sensitivity Analysis,
Knock-out and mutation analysis, Scenario-based analysis

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Pathway Analysis in Systems Biology

Goal: understand how to analyze and explore the behavior of biochemical pathway models.

Topics:

- Pathway databases and model sources (BioModels, KEGG, Reactome)
- Importing and simulating SBML models in Python
- Sensitivity Analysis
- Knock-out and mutation analysis

in Python: `tellurium`, `SALib`, `matplotlib`

PATHWAY DATABASES

Pathway Databases

1. **Descriptive databases:**
   - KEGG: curated maps of metabolic and signaling pathways (`https://www.genome.jp/kegg/`)
   - Reactome: mechanistic descriptions linked to gene/protein data (`https://reactome.org/`)

2. **Quantitative models:**
   - BioModels: repository of mathematical models (ODE/stochastic) in SBML (https://biomodels.org/)
   - Models are curated, documented, and ready for simulation

Example: The EGFR signaling pathway model by Kholodenko (1999)
`BioModels ID: BIOMD0000000048`

Importing an SBML model in Python

Using Tellurium:

- Tellurium supports SBML and Antimony
- Enables simulation, parameter modification, and steady-state analysis

Python example

```
import tellurium as te
r = te.loadSBMLModel('BIOMD0000000048.xml')
result = r.simulate(0, 300, 200)
r.plot()
```

The model can now be used for systematic analysis.

Simulation of SBML models

SMBL models can be simulated either using

- deterministic methods (e.g. CVODE, Runge Kutta), or
- stochastic methods (e.g. Gillespie's SSA).

The choice depends on the system size (e.g., molecules in high/small numbers) and the nature of the reactions (e.g., fast/slow reactions, discrete quantity changes).

It is LEFT TO THE USER to choose the most appropriate method, or to understand for which analysis method the SBML was designed by the authors.

SENSITIVITY ANALYSIS

Sensitivity Analysis – Motivation

Goal: quantify how much EACH PARAMETER influences the system behavior.

Why it matters:

- Identify critical parameters that control dynamics
- Assess robustness (to perturbations) of system behavior and model predictions
- Prioritize parameters for experimental measurement
- Detect potential drug targets

Sensitivity Analysis – Parameters

Which parameters can be analyzed:

- Reaction rate constants (e.g., k_1, k_2)
- Initial concentrations of species

Possibile output metrics:

- Steady-state concentrations of a target species
- Dynamic features (e.g., peak time, amplitude)

Example question: "How sensitive is active ERK concentration to the rate of EGFR activation?"

Sensitivity Analysis – Mathematical Definition

Local sensitivity:

$$S_i(t) = \frac{\partial X(t)}{\partial p_i} \cdot \frac{p_i}{X(t)}$$

- Measures the relative change in output $X(t)$ due to a small change in parameter p_i
- Computed numerically or analytically (when possible)

Global sensitivity:

- Explore the entire parameter space
- Requires sampling and variance-based methods (e.g. Morris, Sobol)

Interpretation: High $S_i \Rightarrow$ parameter p_i strongly influences the model output.

Local vs Global Sensitivity Analysis

Two complementary perspectives:

Local sensitivity:

- Studies small perturbations around a nominal parameter set (typically, at steady-state)
- Assumes parameters are approximately linear near the reference point
- Fast to compute, provides local slopes

Global sensitivity:

- Explores the full parameter space
- Captures nonlinearities and parameter interactions
- Requires multiple simulations (sampling-based)

Analogy: Local $\Rightarrow$ "derivative at a point"; Global $\Rightarrow$ "variance explained by each parameter".

When to use Local vs Global Sensitivity

Local analysis – best when:

- Model is approximately linear near default parameters
- You need quick insight into parameter importance
- You trust your parameter estimates (low uncertainty)

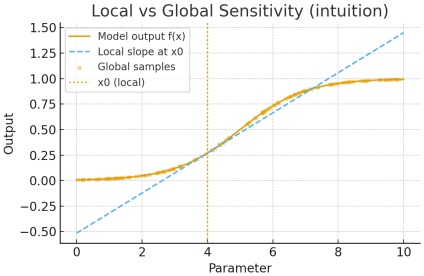
Global analysis – best when:

- Parameters are uncertain or cover large ranges
- The model is nonlinear or exhibits thresholds/switches
- You suspect parameter interactions

Trade-off: Local = cheap, linear, local insight Global = expensive, nonlinear, comprehensive

Sensitivity: Visual Intuition (Local vs Global)

- **Local sensitivity** $\Rightarrow$ derivative at a point: measures the effect of small changes around nominal parameters.
- **Global sensitivity** $\Rightarrow$ explores the entire parameter space: captures *nonlinearities* and *interactions*.
- Practical rule: local = fast and diagnostic; global = more expensive but comprehensive.



Intuition: tangent line (local) vs random samples (global).

Global Sensitivity – Morris Method

Idea: estimate how output changes when each parameter is perturbed along random trajectories in parameter space.

Elementary Effect (EE):

$$EE_i = \frac{f(p_1, \ldots, p_i + \Delta, \ldots, p_k) - f(p_1, \ldots, p_i, \ldots, p_k)}{\Delta}$$

- Compute EE_i for multiple random samples
- Mean of $|EE_i| \rightarrow$ overall influence of parameter i
- Standard deviation of $EE_i \rightarrow$ nonlinearity or interaction effects

Advantages:

- Requires few simulations
- Good screening method for high-dimensional models

Global Sensitivity – Sobol Method

Idea: decompose the total variance of the model output into contributions from each parameter and their interactions.

$$V(Y) = \sum_i V_i + \sum_{i<j} V_{ij} + \dots$$

where V_i is the variance contribution from parameter p_i alone.

Sobol indices:

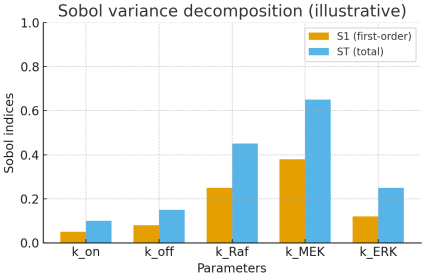
$$S_i = \frac{V_i}{V(Y)} \quad \text{(first-order effect)} \qquad S_{Ti} = \frac{V_i + \text{interactions}}{V(Y)} \quad \text{(total effect)}$$

Interpretation:

- S_i large $\rightarrow$ strong individual effect
- $S_{Ti} - S_i$ large $\rightarrow$ strong interaction with others

Sobol Analysis: Variance Decomposition (S1 vs ST)

- **S1** (first-order): fraction of variance explained by one parameter alone.
- **ST** (total): total contribution of that parameter, including *all interactions*.
- Large gap between ST and S1 $\Rightarrow$ strong *interactions*.
- Common workflow: screen with Morris $\rightarrow$ quantify precisely with Sobol.



Illustrative example: S1 and ST for toy parameters.

Comparing Morris and Sobol Methods

	Morris	**Sobol**
Type	Screening / qualitative	Quantitative / variance-based
Computational cost	Low (hundreds of runs)	High (thousands of runs)
Detects interactions	Yes (via std of EE)	Yes (via total indices S_{Ti})
Output	Rank of parameter importance	Fraction of output variance
Use when	Many uncertain parameters	Precise analysis of few parameters

Practical tip: Use Morris to screen parameters $\Rightarrow$ then refine analysis with Sobol on top candidates.

KNOCKOUT AND MUTATION ANALYSIS

Knock-out and Mutation Analysis – Motivation

Goal: study how the system responds to structural or kinetic perturbations.

Knock-out: Simulate complete loss of a gene or enzyme activity.
Mutation: Simulate partial loss or gain of function (e.g. reduced or increased rate constant).

Applications:

- Predict effects of genetic mutations
- Explore disease mechanisms
- Test hypotheses on pathway robustness

Example: EGFR Pathway and Cancer Mutation

Biological context:

- EGFR is often over-activated in cancer due to mutations
- Leads to excessive ERK phosphorylation $\Rightarrow$ uncontrolled proliferation

Analysis plan:

1. Simulate wild-type pathway (normal EGFR activation)
2. Simulate mutation with k_{on} (EGF binding reaction constant) increased $\times 3$
3. Compare steady-state and dynamics of target proteins

Expected result: Mutant EGFR causes sustained high target proteins even with low EGF stimulus.

Knock-out Example

Hypothesis: Knock-out of MEK should abolish ERK phosphorylation.

Implementation:

- Set rate constant for MEK$\rightarrow$ERK reaction to zero
- Compare time-course of target proteins for wild-type vs knock-out

Result: no ERK activation $\Rightarrow$ MEK is an essential node.

Summary

- Pathway models from BioModels enable reproducible in silico studies
- Sensitivity analysis identifies key regulatory parameters
- Knock-out and mutation simulations reveal functional robustness